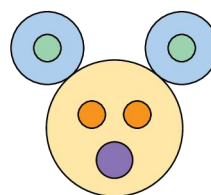


3 points

1. How many circles are there in the figure?

tú zhōng yí gòng yǒu duō shǎo gè yuán
图 中 一 共 有 多 少 个 圆 ?

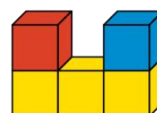


- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

2. The picture shows 5 cubes viewed from the front.

What is the view from above?

yòu tú cóng qián miàn kàn yǒu 5 gè lì fāng tǐ
右 图 从 前 面 看 有 5 个 立 方 体 。



cóng shàng miàn kàn shì shén me yàng zi de
从 上 面 看 是 什 么 样 子 的 ?

- (A) (B) (C) (D) (E)

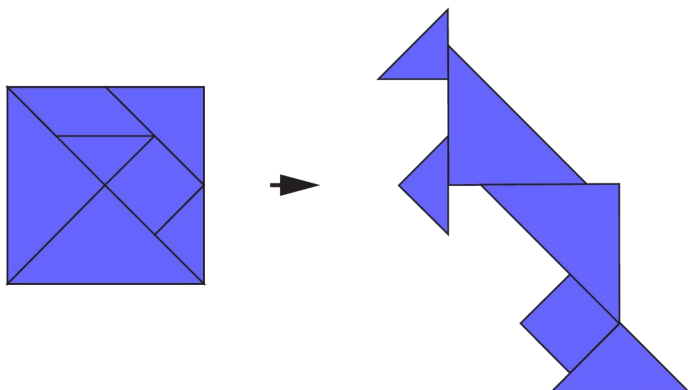
3. Each bowl contains four numbered balls, as shown. In which bowl is the sum of all the numbers largest?

xià miàn měi gè wǎn lǐ dōu yǒu sì kē shù zì qiú, nǎ gè wǎn lǐ de shù zì zhī hé zuì dà
下 面 每 个 碗 里 都 有 四 颗 数 字 球, 哪 个 碗 里 的 数 字 之 和 最 大 ?

- (A) (B) (C) (D) (E)

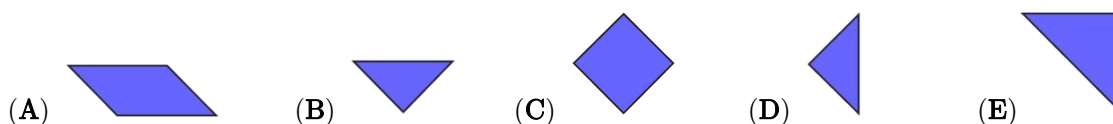
4. Mr Beaver rearranges the pieces to make a kangaroo figure.

xiǎo wēi yòng qī qiǎo bǎn pīn yì zhī dài shǔ
小 威 用 七 巧 板 拼 一 只 袋 鼠 。



Which piece is missing?

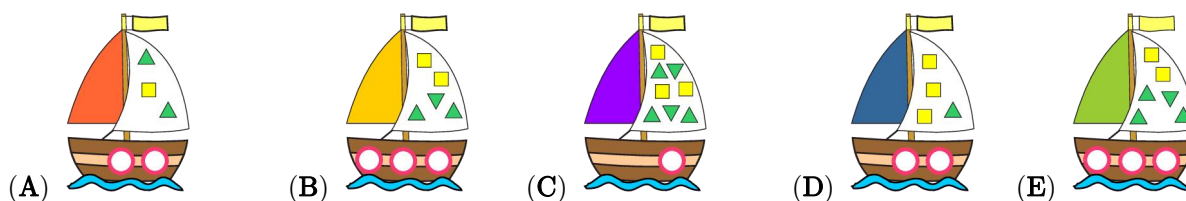
hái quē xià miàn nǎ yí kuài
还 缺 下 面 哪 一 块 ？



5. My boat has more than 1 circle. It also has 2 more triangles than squares. Which boat is mine?

wǒ de chuán shàng yǒu bù zhǐ 1 gè yuán, sān jiǎo xíng bǐ zhèng fāng xíng duō 2 gè。下 面 哪
我的 船 上 有 不 止 1 个 圆，三 角 形 比 正 方 形 多 2 个。

tiáo chuán shì wǒ de
条 船 是 我 的 ？



6. This is my grandfather's birthday cake. A large candle stands for 10 years and a small one for 1 year. How old is my grandfather?

zhè shì yé ye de shēng rì dàn gāo yì gēn cháng là zhú dài biǎo suì yì gēn
这是爷爷的生日蛋糕。一根长蜡烛代表 10 岁，一根

duǎn là zhú dài biǎo suì
短蜡烛代表 1 岁。

yé ye duō dà le
爷爷多大了？

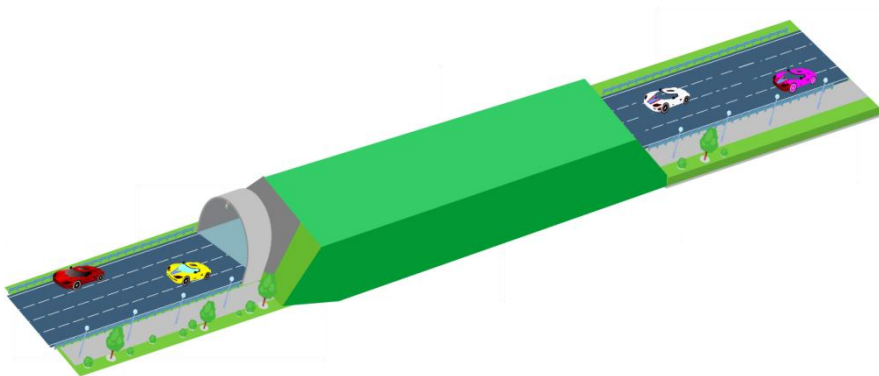


- (A) 65 (B) 66 (C) 76 (D) 77 (E) 78

7. Pablo puts 10 toy cars on this racetrack. How many cars are in the tunnel?

xiǎo bù zài zhè gè sài dào shàng yì gòng fàng le liàng wán jù chē suì dào lǐ yǒu duō shǎo
小布在这个赛道上一共放了 10 辆玩具车，隧道里有多少

liàng wán jù chē
辆玩具车？



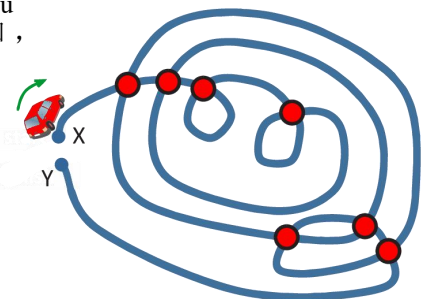
- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

8. Steven drives from X to Y. At each crossing, he stops before going straight ahead. In total, how many times does he stop at a crossing?

wén wén kāi chē cóng X chéng dào Y chéng zài měi gè shí zì lù kǒu
文文开车从 X 城到 Y 城，在每个十字路口，

tā dōu huì tíng xià lái rán hòu zài zhí xíng tā zǒng gòng zài shí
他都会停下来，然后再直行。他总共在十

zì lù kǒu tíng le duō shǎo cì
十字路口停了多少次？



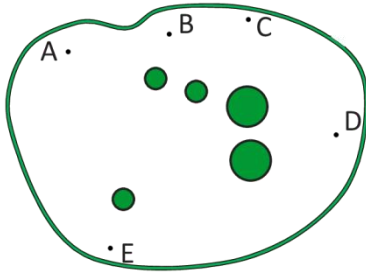
- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

4 points

9. There are 5 trees in a park. A beaver can see only two of the trees because all the others are hidden behind other trees. At which of the marked points is the beaver standing?

yí gè gōng yuán lǐ yǒu 5 kē shù yì zhī hǎi lí zhǐ néng kàn dào qí zhōng liǎng kē shù yīn wéi
一个公园里有5棵树。一只海狸只能看到其中两棵树，因为

qí tā shù dōu bèi zhè liǎng kē shù dǎng zhù le hǎi lí zhàn zài tú zhōng nǎ gè diǎn shàng ?
其他树都被这两棵树挡住了。海狸站在图中哪个点上？



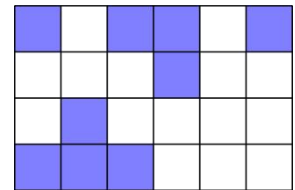
(A) at A A 点 (B) at B B 点 (C) at C C 点 (D) at D D 点 (E) at E E 点

10. There are 24 squares in the picture. Suchit has coloured some of the squares. How many more squares need to be coloured so that half of the squares are coloured?

tú zhōng yǒu 24 gè fāng kuài sū sū yǐ jīng gěi yì xiē fāng kuài tú shàng
图中有24个方块。苏苏已经给一些方块涂上

le yán sè hái xū yào zài gěi duō shǎo gè fāng kuài shàng sè cái néng
了颜色，还需要再给多少个方块上色，才能

shǐ dé yí bàn de fāng kuài dōu yǒu yán sè ?
使得一半的方块都有颜色？



(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

11. The two tokens with the question mark have the same number. What is each missing number so that the sum is 18?

liǎng gè dài wèn hào de yuán quān lǐ yǒu xiāng tóng de shù zì wèi le shǐ zǒng hé wéi 18 měi
两个带问号的圆圈里有相同的数字。为了使总和为18，每

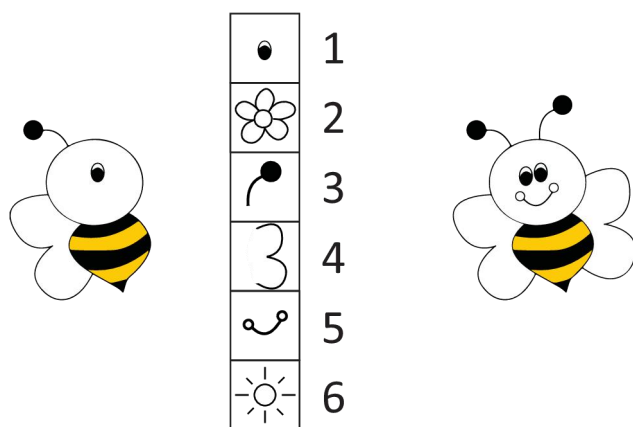
gè wèn hào yuán quān lǐ de shù zì shì duō shǎo ?
个问号圆圈里的数字是多少？

$$\textcircled{10} + \textcircled{?} + \textcircled{?} + \textcircled{2} = 18$$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

12. Raha wants to finish the bee on the left according to the model on the right.

xiǎo lā xiǎng àn zhào yòu biān de tú bǎ zuǒ biān de mì fēng bǔ chōng wán zhěng
小拉想按照右边的图把左边的蜜蜂补充完整。



Raha needs to win points to unlock parts of the bee. How many points does she need to win to complete the bee?

xiǎo lā xū yào yíng dé jī fēn lái jiě suǒ mì fēng de gè gè shēn tǐ bù wèi tā hái xū yào yíng
小拉需要赢得积分来解锁蜜蜂的各个身体部位。她还需要赢

dé duō shǎo fēn cái néng wán chéng zhè zhī mì fēng
得多少分才能完成这只蜜蜂？

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

13. The table has 30 boxes. After painting the boxes in row 3, row 6, column C and column D, how many boxes will be not painted?

biǎo gé lǐ yǒu 30 gè gé zǐ. gěi dì 3 háng dì 6 háng dì liè hé dì liè de gé zǐ tú sè
表格里有 30 个格子。给第 3 行、第 6 行、第 C 列和第 D 列的格子涂色

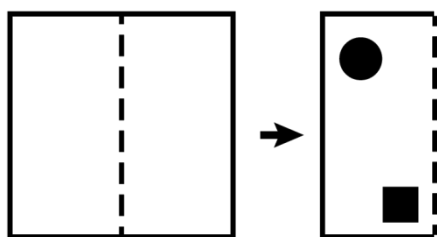
hòu hái yǒu duō shǎo gè gé zǐ méi yǒu bèi tú sè
后，还有多少个格子没有被涂色？

	A	B	C	D	E
1					
2					
3					
4					
5					
6					

- (A) 8 (B) 10 (C) 12 (D) 18 (E) 22

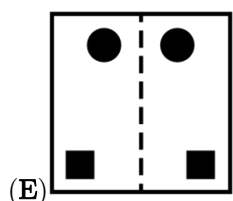
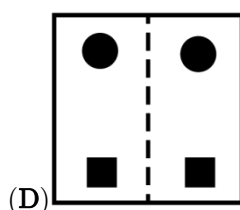
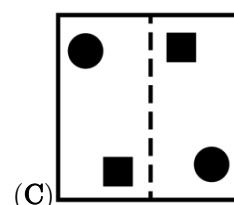
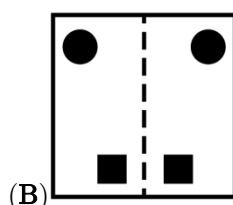
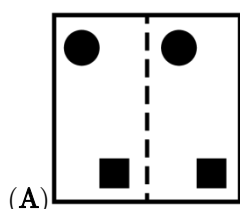
14. A sheet of paper is folded in half. Square and round holes are punched.

jiāng yì zhāng zhǐ duì zhé hòu , dǎ shàng fāng xíng kǒng hé yuán xíng kǒng 。
将 一 张 纸 对 折 后 ， 打 上 方 形 孔 和 圆 形 孔 。



How does the sheet look after it is unfolded again?

zhǎn kāi hòu zhè zhāng zhǐ shì shén me yàng zi de ?
展 开 后 这 张 纸 是 什 么 样 子 的 ？

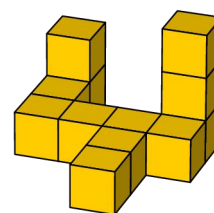


15. A student made the shape shown using 12 cubes. He put one drop of glue between any two cubes that share a common face. How many drops of glue did he use?

yí gè xué shēng yòng 12 gè lì fāng tǐ dā chéng le tú zhōng de xíng zhuàng 。
一 个 学 生 用 12 个 立 方 体 搭 成 了 图 中 的 形 状 。

gòng yòng yí gè miàn de liǎng gè lì fāng tǐ zhī jiān tā dōu yòng yì dī
共 用 一 个 面 的 两 个 立 方 体 之 间 ， 他 都 用 一 滴

jiāo shuǐ zhān zhù , tā yí gòng yòng le duō shǎo dī jiāo shuǐ
胶 水 粘 住 ， 他 一 共 用 了 多 少 滴 胶 水 ？



(A) 8

(B) 9

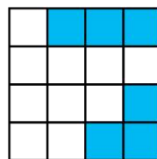
(C) 10

(D) 11

(E) 12

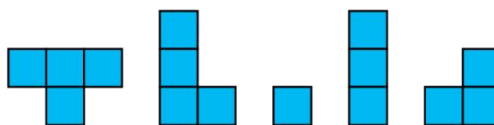
16. Max wants to complete the puzzle shown.

xiǎo mài xiǎng wán chéng tú zhōng de pīn tú
小麦想完成图中的拼图。



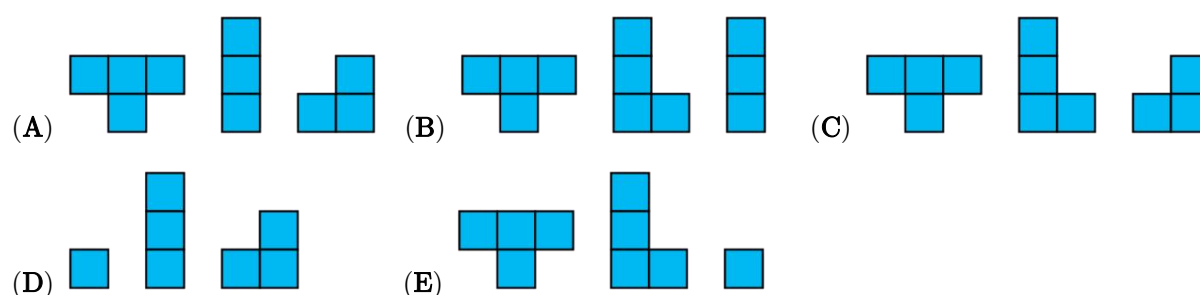
He has 5 different pieces shown.

tā yǒu gè bù tóng de pīn tú kuài
他有 5 个不同的拼图块。



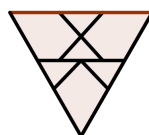
Which pieces does he have to use to complete the puzzle?

tā bì xū yòng nǎ jǐ kuài lái wán chéng zhè gè pīn tú
他必须用哪几块来完成这个拼图？



5 points

17. Elvis has 6 identical triangles like this.

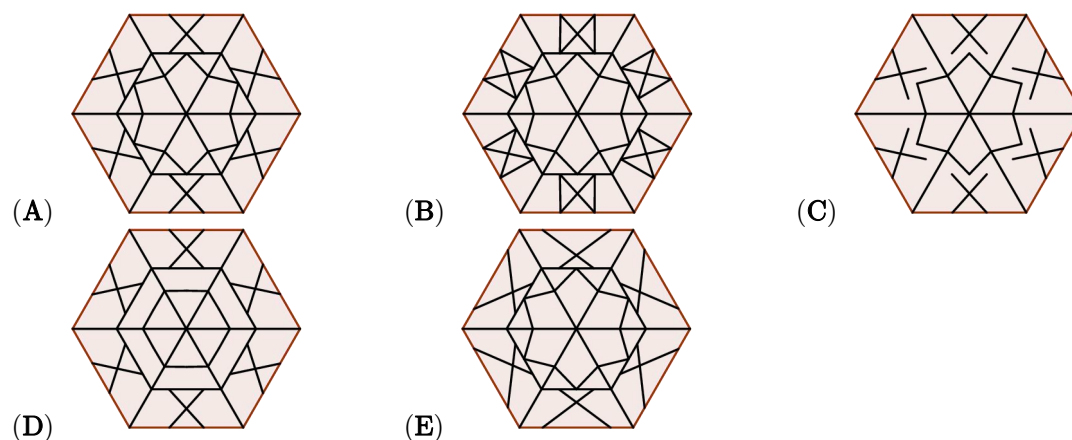


Which of the following pictures can he make?

yī yī yǒu gè zhè yàng de
依依有 6 个这样的

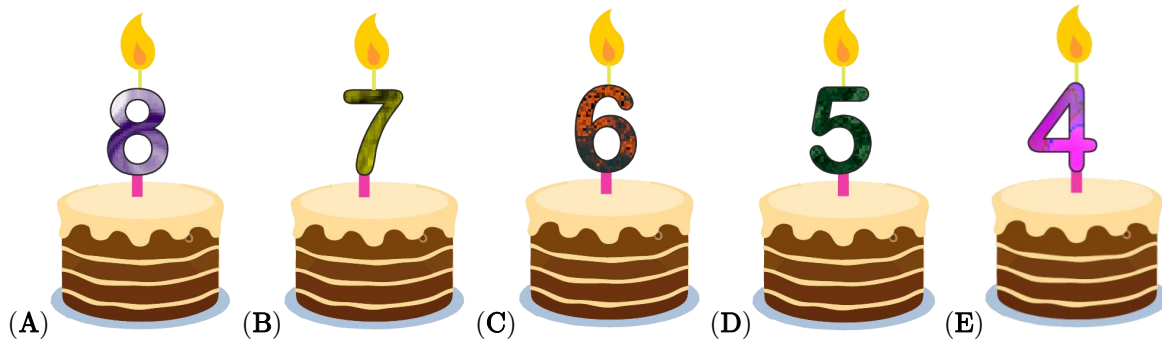


sān jiǎo xíng, tā néng pīn chū xià miàn nǎ gè tú àn
三角形, 他能拼出下面哪个图案?



18. Five children share a birthday and each child has their own cake. Lea is two years older than Jose, but one year younger than Ali. Vittorio is the youngest. Which is Sarah's cake?

wǔ gè hái zi yì qǐ guò shēng rì měi gè hái zi dōu yǒu zì jǐ de dàn gāo xiǎo lǐ bǐ xiǎo sī dà liǎng suì dàn bǐ ā lǐ xiǎo yí suì xiǎo wéi shì nián líng zuì xiǎo de nǎ gè shì shā shā de dàn gāo ?
五个孩子一起过生日，每个孩子都有自己的蛋糕。小李比小斯大两岁，但比阿里小一岁。小维是年龄最小的。哪个是莎莎的蛋糕？

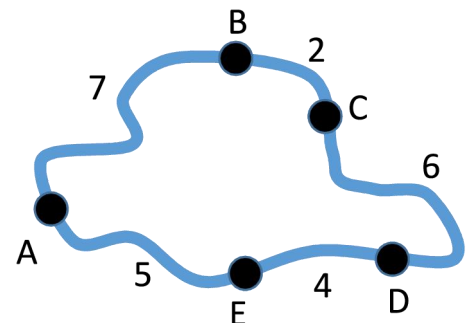


19. The map shows five villages A, B, C, D and E, and the distances in kilometres between them. Only two villages are the same distance apart no matter which route you choose. Which are these two villages?

dì tú shàng xiǎn shì le wǔ gè cūn zhuāng A、B、C、D 和 E 以及它们之间的距离（以千米为单位）。其中只有两个村庄，无论你选择哪条路线，这

两个村庄之间的距离都是相等的。

qǐng wèn shì nǎ liǎng gè cūn zhuāng ?
请问是哪两个村庄？

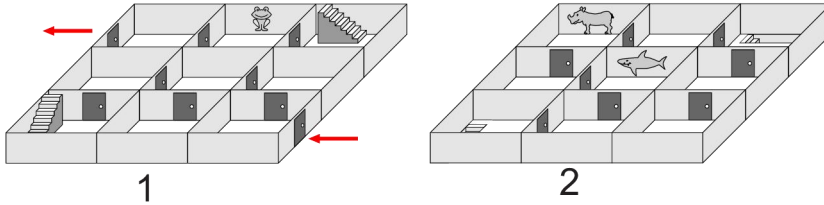


- (A) B and E B 和 E (B) B and D B 和 D (C) C and E C 和 E
(D) A and C A 和 C (E) A and D A 和 D

20. Sam walks through a two-storey maze from the entrance to the exit, both located at floor 1. In what order will she find the wall stickers?

xiǎo sà cóng lóu rù kǒu zǒu dào lóu chū kǒu xū yào chuān guò yí gè liǎng céng lóu de mí
小萨从1楼入口走到1楼出口，需要穿过一个两层楼的迷

宫。她会按照什么顺序看到墙上的贴纸？



- (A) (B) (C) (D) (E)

21. Emma finished third in a solo dance competition. There were three dancers between her and last place. In total, how many dancers took part in the competition?

xiǎo ài zài gè rén wǔ dǎo bǐ sài zhōng huò dé dì sān míng tā hé zuì hòu yí míng zhī jiān yǒu
小艾在个人舞蹈比赛中获得第三名，她和最后一名之间有

sān míng xuǎn shǒu zǒng gòng yǒu duō shǎo míng xuǎn shǒu cān jiā le bǐ sài
三名选手。总共有多少名选手参加了比赛？

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

22. Malik places one of the five pieces on the grid. He cannot rotate or flip the pieces.

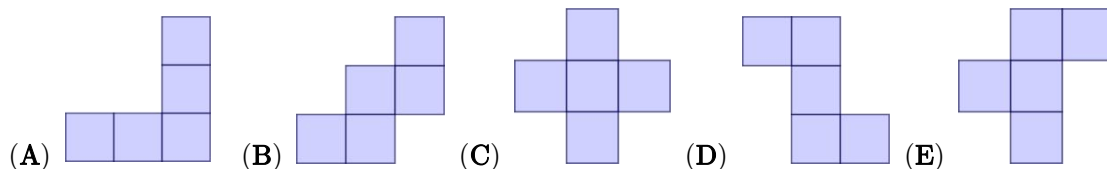
Which piece should he use to cover the numbers with the largest sum?

xiǎo mǎ xiǎng jiāng wǔ gè xuǎn xiàng zhōng de yí gè pīn tú kuài fàng zài yòu
小马想将五个选项中的一个拼图块放在右

cè wǎng gé shàng tā bù néng xuán zhuǎn huò fān zhuǎn zhè xiē pīn tú kuài
侧网格上，他不能旋转或翻转这些拼图块。

1	6	7
9	5	4
2	8	3

tā yòng nǎ gè pīn tú kuài fù gài de shù zì zǒng hé zuì dà
他用哪个拼图块覆盖的数字总和最大？



23. Three frogs live in a pond. Each night, one of the frogs sings a song to the other two. After 9 nights, one of the frogs had sung 2 times. Another frog had listened to 5 songs. How many songs had the third frog listened to?

sān zhī qīng wā zhù zài yí gè chí táng lǐ měi tiān wǎn shàng qí zhōng yì zhī qīng wā dōu huì
三 只 青 蛙 住 在 一 个 池 塘 里 。 每 天 晚 上 ， 其 中 一 只 青 蛙 都 会
gěi lìng wài liǎng zhī qīng wā chàng yì shǒu gē gè wǎn shàng zhī hòu qí zhōng yì zhī qīng wā
给 另 外 两 只 青 蛙 唱 一 首 歌 。 9 个 晚 上 之 后 ， 其 中 一 只 青 蛙
chàng le 2 cì gē lìng yì zhī qīng wā tīng le 5 shǒu gē dì sān zhī qīng wā tīng le duō shǎo
唱 了 2 次 歌 ， 另 一 只 青 蛙 听 了 5 首 歌 ， 第 三 只 青 蛙 听 了 多 少
shǒu gē
首 歌 ？

- (A) 7 (B) 6 (C) 5 (D) 4 (E) 3

24. Digits 1, 1, 2 and 3 are printed on four different cards. Three cards are laid out to make a subtraction, as shown in the picture. How many different results can be obtained?



shù zì 1, 1, 2 hé 3 bèi dǎ yìn zài sì zhāng kǎ piàn shàng jiāng sān zhāng kǎ piàn bǎi zài yì qǐ
数 字 1、1、2 和 3 被 打 印 在 四 张 卡 片 上 。 将 三 张 卡 片 摆 在 一 起
zuò jiǎn fǎ rú tú suǒ shì kě yǐ dé dào duō shǎo zhǒng bù tóng de jié guǒ
做 减 法 ， 如 图 所 示 。 可 以 得 到 多 少 种 不 同 的 结 果 ？

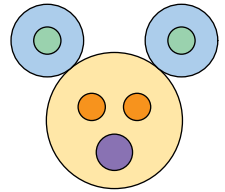
- (A) 6 (B) 8 (C) 10 (D) 12 (E) 24

Pre-Ecolier

3 points

1 (Myanmar). How many circles are there in the figure?

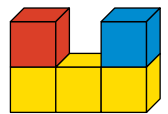
- (A) 5
(B) 6
(C) 7
(D) 8
(E) 9



SOLUTION: There are 3 outer circles and 5 inner circles, so in total 8 circles.

2 (Myanmar). The picture shows 5 cubes viewed from the front. What is the view from above?

- (A) (B) (C) (D) (E)



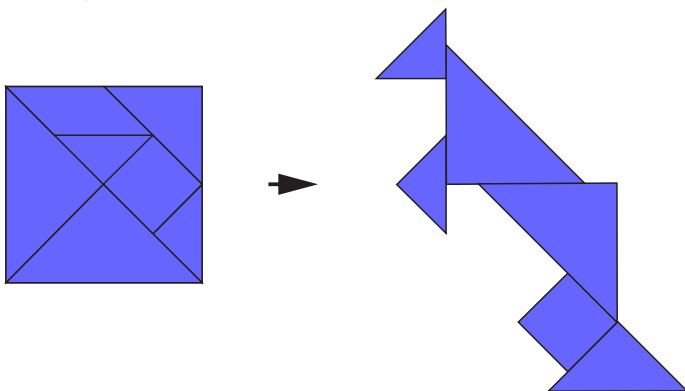
SOLUTION: The cubes on the top are left red and blue right. The middle doesn't have a cube on the top, so the colour you see is yellow. Therefore, from left to right, we see red, yellow, blue.

3 (Croatia). Each bowl contains four numbered balls, as shown. In which bowl is the sum of all the numbers largest?

- (A) (B) (C) (D) (E)

SOLUTION: Notice that in each bowl there is a ball with number 9, 7 and 4. So the sum depends only on the fourth number and is the largest in A.

4 (China). Mr Beaver rearranges the pieces to make a kangaroo figure.

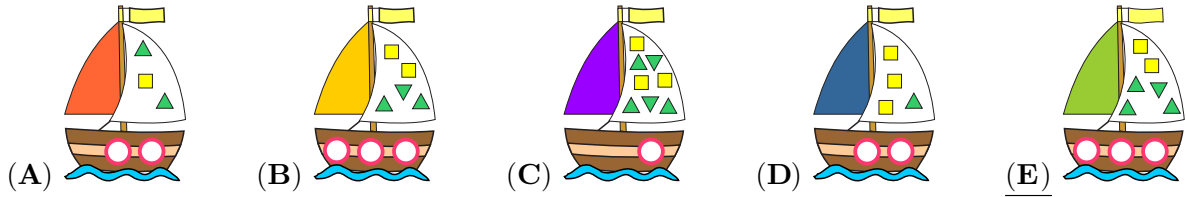


Which piece is missing?

- (A) (B) (C) (D) (E)

SOLUTION: The kangaroo is made of 5 triangles and 1 square. So Mr Beaver did not use the piece in option A.

5 (Slovakia). My boat has more than 1 circle. It also has 2 more triangles than squares. Which boat is mine?



SOLUTION: My boat has more than 1 circle, therefore we reject the answer C. Now we count how many triangles and squares are on the sails. In A there are 2 triangles and 1 square (1 more triangle than square) - not the correct answer. In B there are 3 triangles and 2 squares (1 more triangle than squares) - not the correct answer. In D there are 1 triangle and 3 squares (2 less triangle than squares) - not the correct answer. In E there are 4 triangles and 2 squares (2 more triangle than squares), so this is my boat.

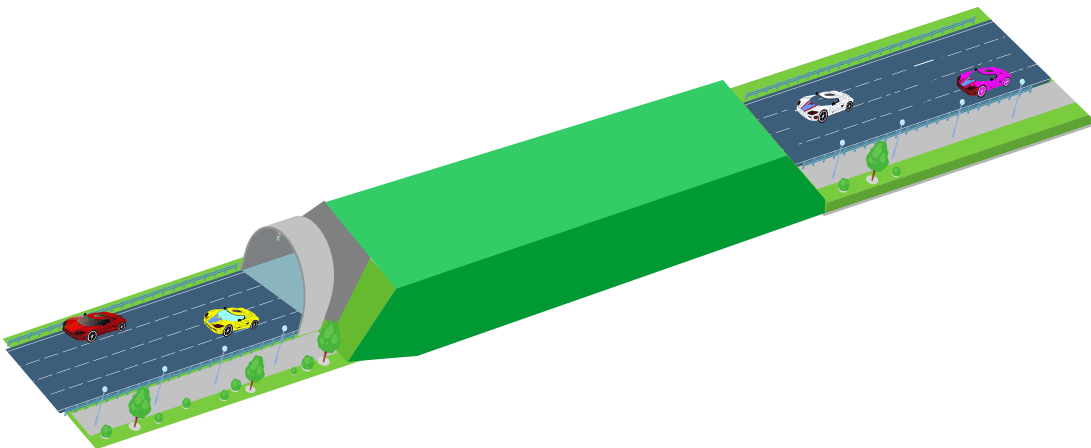
6 (France). This is my grandfather's birthday cake. A large candle stands for 10 years and a small one for 1 year. How old is my grandfather?

- (A) 65 (B) 66
(C) 76 (D) 77
(E) 78



SOLUTION: There are 7 big candles and 6 small ones on the cake. So grandfather is $7 \times 10 + 6 \times 1 = 76$ years old.

7 (Montenegro). Pablo puts 10 toy cars on this racetrack. How many cars are in the tunnel?

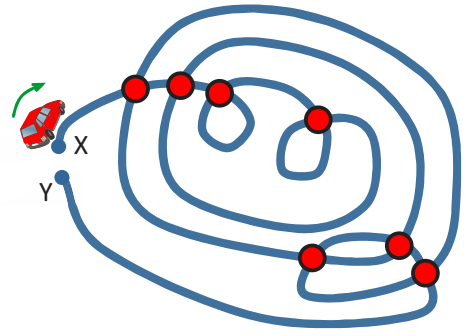


- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

SOLUTION: There are in total 10 cars on the racetrack. We can see 4 cars that are not in the tunnel, so $10 - 4 = 6$ cars are in the tunnel.

8 (Greece). Steven drives from X to Y. At each crossing, he stops before going straight ahead. In total, how many times does he stop at a crossing?

- (A) 11
(B) 12
(C) 13
(D) 14
(E) 15

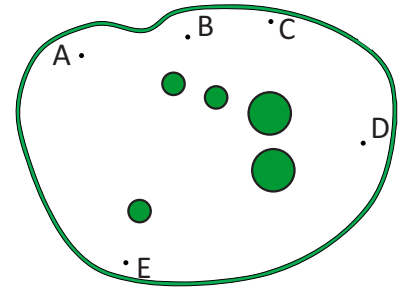


SOLUTION: The car will stop each time it reaches a red dot, and then continue straight ahead. All of the crossings will be passed exactly twice. In total, the car will stop 14 times.

4 points

9 (Greece). There are 5 trees in a park. A beaver can see only two of the trees because all the others are hidden behind other trees. At which of the marked points is the beaver standing?

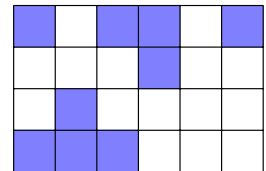
- (A) at A
(B) at B
(C) at C
(D) at D
(E) at E



SOLUTION: From any point except D, the beaver can see more than 2 trees. From the point D, the two big trees block all rest from view.

10 (Canada). There are 24 squares in the picture. Suchit has coloured some of the squares. How many more squares need to be coloured so that half of the squares are coloured?

- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5



SOLUTION: Half the number of squares is $24/2 = 12$, and 9 squares are already coloured. Therefore, 3 more need to be coloured.

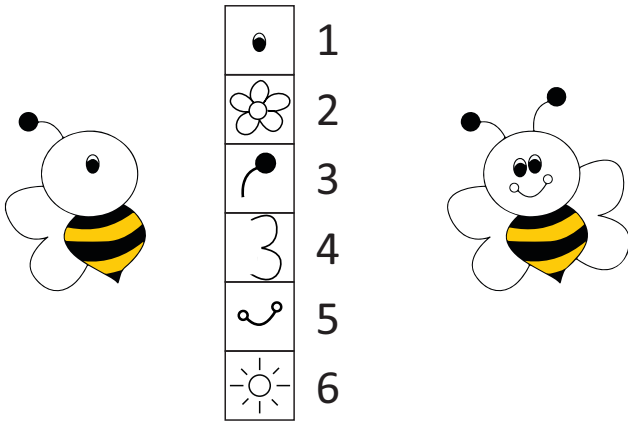
11 (Denmark). The two tokens with the question mark have the same number. What is each missing number so that the sum is 18?

$$\textcircled{10} + \textcircled{?} + \textcircled{?} + \textcircled{2} = 18$$

- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5

SOLUTION: We can add $10 + 2 = 12$. Then $18 - 12 = 6$ and so the number on each token is 3.

12 (Iran). Raha wants to finish the bee on the left according to the model on the right.



Raha needs to win points to unlock parts of the bee. How many points does she need to win to complete the bee?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

SOLUTION: The missing part of the bee are 1 eye, 1 antenna, 1 wing, 1 mouth. Altogether these need $1 + 3 + 4 + 5 = 13$ points.

13 (Russia). The table has 30 boxes. After painting the boxes in row 3, row 6, column C and column D, how many boxes will be not painted?

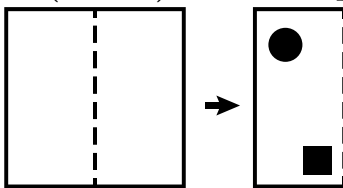
	A	B	C	D	E
1					
2					
3					
4					
5					
6					

- (A) 8 (B) 10 (C) 12 (D) 18 (E) 22

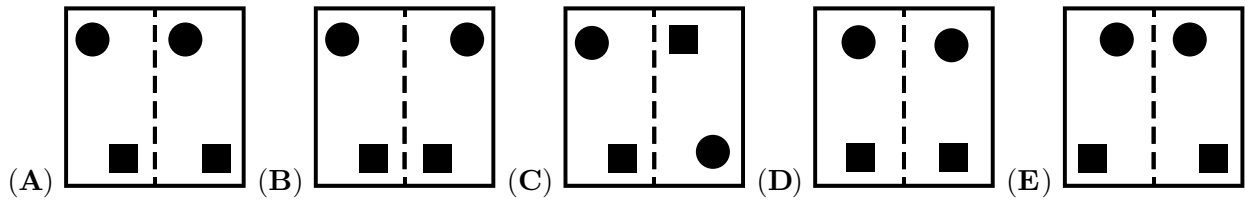
SOLUTION: We can mark the squares and count how many are left.

	A	B	C	D	E
1					
2					
3					
4					
5					
6					

14 (Russia). A sheet of paper is folded in half. Square and round holes are punched.



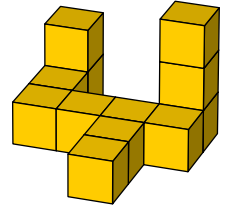
How does the sheet look after it is unfolded again?



SOLUTION: After unfolding, the two square holes will be closer together than the two round holes.

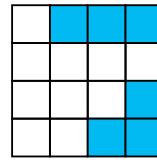
15 (Greece). A student made the shape shown using 12 cubes. He put one drop of glue between any two cubes that share a common face. How many drops of glue did he use?

- (A) 8 (B) 9 (C) 10
(D) 11 (E) 12

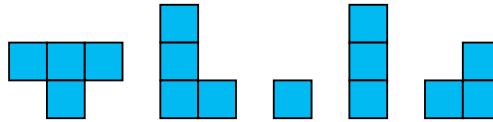


SOLUTION: By counting carefully, we see that there must be 11 drops of glue.

16 (Poland). Max wants to complete the puzzle shown.



He has 5 different pieces shown.



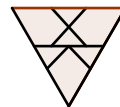
Which pieces does he have to use to complete the puzzle?

- (A) (B) (C)
(D) (E)

SOLUTION: He has to use 1 at the top, 4 at the right and 5 at the bottom.

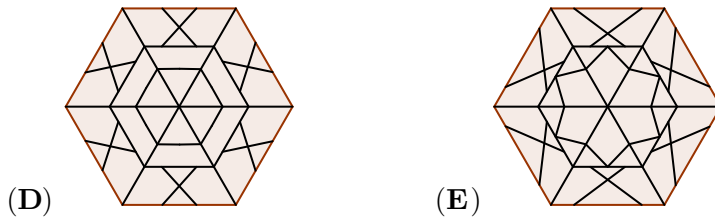
5 points

17 (Denmark). Elvis has 6 identical triangles like this.



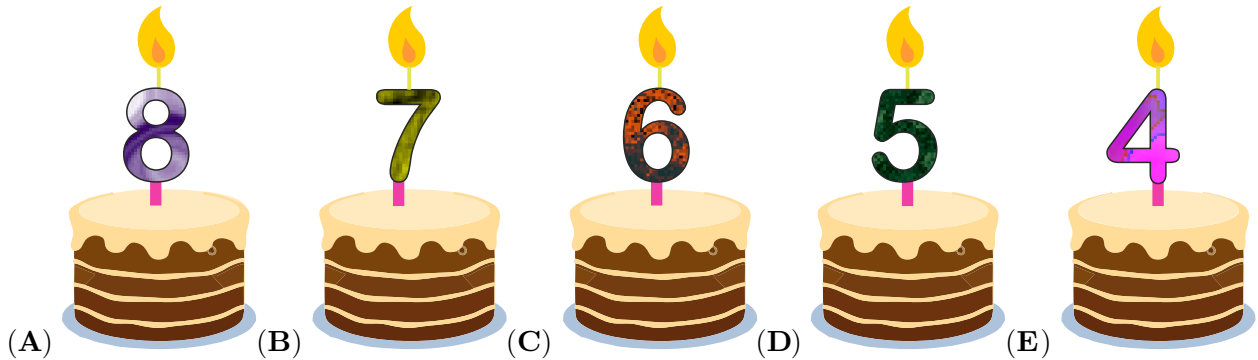
Which of the following pictures

- (A) (B) (C)



SOLUTION: We only need to compare one piece of each option and notice that only A is correct.

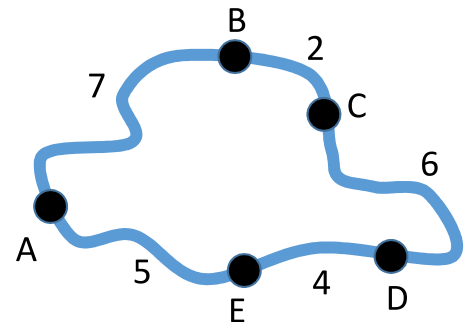
18 (Poland). Five children share a birthday and each child has their own cake. Lea is two years older than Jose, but one year younger than Ali. Vittorio is the youngest. Which is Sarah's cake?



SOLUTION: The five children are aged 8, 7, 6, 5 and 4. Vittorio is the youngest so he is 4. Lea is two years older than Jose and one year younger than Ali, so Jose is 5, Lea is 7 and Ali is 8 years old. So the fifth child, Sarah, is 6 years old.

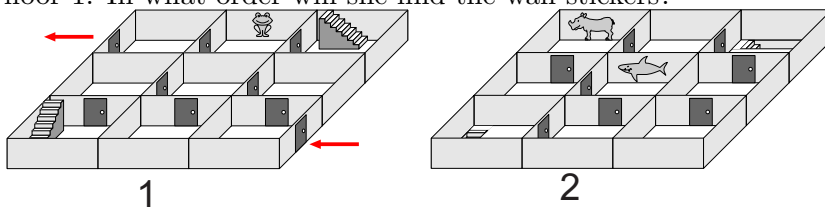
19 (Greece). The map shows five villages A, B, C, D and E, and the distances in kilometres between them. Only two villages are the same distance apart no matter which route you choose. Which are these two villages?

- (A) B and E (B) B and D
(C) C and E (D) A and C
(E) A and D



SOLUTION: The whole circuit is $7 + 2 + 6 + 4 + 5 = 24$ km. So if the two distances are equal, each must be 12 km. So we are looking for a village that there is another one at a distance 12 km going, say, clockwise. From A, we have B is distance 7 km (too little), C is $7 + 2 = 9$ km (also too little) and D is $7 + 2 + 6 = 15$ km (too much). So A is not one of the villages. Try B. We have distances 2, $2 + 6 = 8$, $2 + 6 + 4 = 12$ km. This is acceptable, so the two villages are B and E. One can verify that the other direction joining B and E is 12 km, as before. One can also verify that the other cases, such as from C or D, do not give us distance 12 km to any village, so they are not the answer.

20 (Belgium). Sam walks through a two-storey maze from the entrance to the exit, both located at floor 1. In what order will she find the wall stickers?



- (A) (B) (C) (D) (E)

SOLUTION: There is only one route from the entrance to the exit. In order, Sam will see the shark,

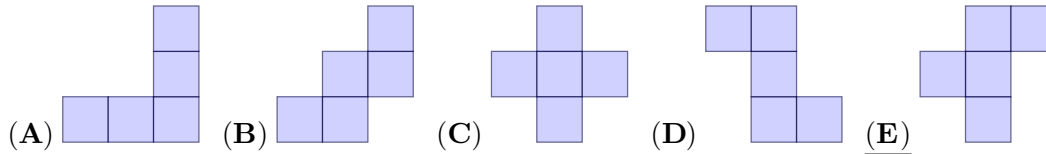
rhinoceros and frog.

21 (Slovakia). Emma finished third in a solo dance competition. There were three dancers between her and last place. In total, how many dancers took part in the competition?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

SOLUTION: Emma finished third, so there are at least 3 dancers in the competition, including her. After Emma we have 3 more dancers and the dancer who is in last place. In total that will be $3 + 3 + 1 = 7$ dancers in the competition.

22 (Canada). Malik places one of the five pieces on the grid. He cannot rotate or flip the pieces. Which piece should he use to cover the numbers with the largest sum?



1	6	7
9	5	4
2	8	3

SOLUTION: All pieces cover five squares on the grid. The five largest numbers are 9, 8, 7, 6 and 5. Only one piece will cover all of these.

23 (Lithuania). Three frogs live in a pond. Each night, one of the frogs sings a song to the other two. After 9 nights, one of the frogs had sung 2 times. Another frog had listened to 5 songs. How many songs had the third frog listened to?

- (A) 7 (B) 6 (C) 5 (D) 4 (E) 3

SOLUTION: On each of the 9 nights, one frog sang, the other two listened. The first frog sang 2 times; the second frog listened 5 times. This means the third frog sang 3 times, and so it must have listened $9 - 3 = 6$ times.

24 (Hong Kong). Digits 1, 1, 2 and 3 are printed on four different cards. Three cards are laid out to make a subtraction, as shown in the picture. How many different results can be obtained?



- (A) 6 (B) 8 (C) 10 (D) 12 (E) 24

SOLUTION: With digits 1, 1, 2 and 3 we can make expressions: $11 - 2 = 9$, $11 - 3 = 8$, $12 - 1 = 11$, $12 - 3 = 9$, $13 - 1 = 12$, $13 - 2 = 11$, $21 - 1 = 20$, $21 - 3 = 18$, $23 - 1 = 22$, $31 - 1 = 30$, $31 - 2 = 28$, $32 - 1 = 31$. There are 12 expressions but only 10 unique results.